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DD FORM 1426 6721 (COORDINATED)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

5310-1570

P.A. NAVY-AS
Other Govt ARMY-AV AIR FORCE-11
INTERNATIONAL INTEREST
PROCUREMENT SPECIFICATION MIL-N-25027
SUPERSEDES NAS 686, NAS 1031 (IN PART)

(F) ENTIRE STANDARD REVISED

TITLE NUT, SELF-LOCKING, PLATE, TWO LUG, FLOATING, LOW HEIGHT, CRES, 125 KSI FTU, 450° F & 800° F

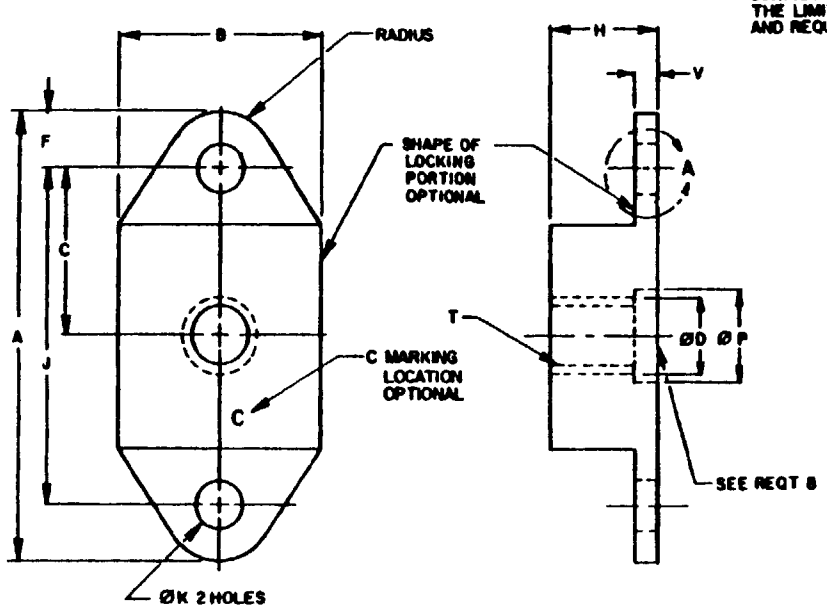
MILITARY STANDARD

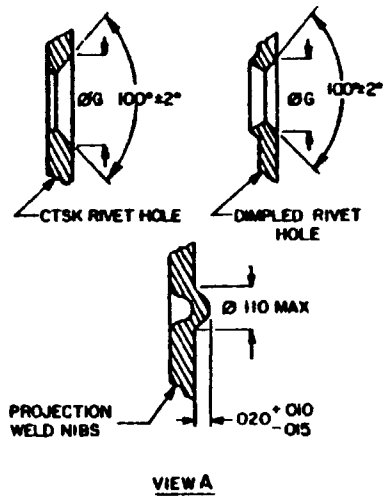
MS 21060

PAGE 1 OF 5

FED SUP CLASS 5310

CONFIGURATION OF NUT OPTIONAL WITHIN THE LIMITATIONS IMPOSED BY DIMENSIONS AND REQUIREMENTS AS SPECIFIED HEREIN.





VIEW A

TABLE I DASH NUMBERS AND DIMENSIONS

DASH NUMBERS 450°F PLAIN HOLES	450°F CSK OR DIMPLED HOLES	450°F WELDING PROJECTIONS	800°F PLAIN HOLES	800°F CSK OR DIMPLED HOLES	800°F WELDING PROJECTIONS	T THREAD SIZES	A MAX	B MAX	B MIN	C ±.005	ØD MIN	F MIN	ØG ±.010	H MAX	J ±.002	Ø E ±.005 -.000	Ø P MAX	Ø P MIN	V MAX	AXIAL STRENGTH LBS/MIN	WEIGHT LBS/100
L04	L04K	L04W	-04	-04K	-04W	112-40 UNJC-3B	.940	.616	.290	.344	—	100	200	175	.680	.098	.270	.172	.032	750	30
L06	L06K	L06W	-06	-06K	-06W	138-32 UNJC-3B	.940	.616	.290	.344	—	100	200	203	.680	.098	.270	.198	.032	1,130	39
L08	L08K	L08W	-08	-08K	-08W	164-32 UNJC-3B	.940	.616	.290	.344	.160	100	200	250	.680	.098	.270	.224	.032	1,720	60
L3	L3K	L3W	-3	-3K	-3W	190-32 UNJF-3B	.940	.616	.290	.344	.194	.100	200	250	.680	.098	.270	.250	.032	2,460	61
L4	L4K	L4W	-4	-4K	-4W	250-28 UNJF-3B	1.292	.516	.350	.500	.254	100	200	.201	1.000	.098	.330	.310	.032	4,500	90
L5	L5K	L5W	-5	-5K	-5W	.3125-24 UNJF-3B	1.292	.609	.412	.500	.317	125	230	320	1.000	.130	.392	.372	.045	7,390	1 26
L6	—	L6W	-6	—	-6W	.375-24 UNJF-3B	1.292	.680	.475	.500	.379	125	230	344	1.000	.130	.455	.435	.055	11,050	1 55
L7	—	L7W	-7	—	-7W	.4375-20 UNJF-3B	1.677	.719	.672	.562	.442	166	230	504	1.125	.161	.517	.497	.115	15,450	2 30
L8	—	L8W	-8	—	-8W	.500-20 UNJF-3B	1.602	.859	.796	.625	.504	166	230	630	1.250	.161	.580	.560	.115	21,110	4 00

1/ MINIMUM "H" NOT SPECIFIED, LIMITED ONLY BY STRENGTH REQUIREMENTS OF SPECIFICATION
2/ DIMPLED RIVET HOLE TOLERANCE FOR Ø "E" IS +.015 -.000

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REQUIREMENTS:

1. MATERIAL: CORROSION RESISTANT STEEL, TYPE A286, (UNS S66286) IN ACCORDANCE WITH AMS5525, AMS5732 OR AMS5737.

2. FINISH: ONLY 800°F NUTS: SILVER PLATE ALL THREADED SURFACES OF NUT ELEMENT IN ACCORDANCE WITH AMS 2410. THREADS SHALL SHOW COMPLETE COVERAGE. SILVER PLATING ON ALL OTHER SURFACES OPTIONAL. TARNISHING OR DISCOLORATION OF SILVER NOT CAUSE FOR REJECTION.

3. LUBRICANT: ONLY 450°F NUTS: DRY FILM LUBRICANT IN ACCORDANCE WITH MIL-M-25027.

4. DIMENSIONING AND TOLERANCING: DIMENSIONING AND TOLERANCING SHALL BE IN ACCORDANCE WITH ANSI Y14.5M.

5. HARDNESS: 49HRC, MAX.

6. THREADS: THREADS BEFORE LUBRICATION IN ACCORDANCE WITH MIL-S-8879.

7. SURFACE TEXTURE: SURFACE TEXTURE, UNLESS OTHERWISE SPECIFIED, SHALL NOT EXCEED 125 MICRODINCHES, IN ACCORDANCE WITH ANSI B46.1.

8. COUNTERBORE/COUNTERSINK: ON SIZE .164 AND LARGER, THREAD RELIEF SHALL BE .062 MINIMUM; ON SIZE .138 AND SMALLER, COUNTERSINK OR RADIUSED WITHIN "F" DIAMETER.

9. PERFORMANCE: SEE PROCUREMENT SPECIFICATION, EXCEPT 450°F DRY FILM LUBRICATED NUTS LIMITED TO FIVE HEAT CYCLES.

10. FLOAT OF NUT POSITION: FLOAT OF NUT POSITION OF ASSEMBLY SHALL NOT BE LESS THAN .030 RADIALLY FROM CENTERED POSITION. NUT ELEMENT SHALL BE CAPABLE OF ENGAGEMENT WITH BOLT IN THE MAXIMUM MISALIGNED POSITION

11. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I.

EXAMPLE: MS21060L4K

- L4K — DASH NUMBER
- MS21060 — BASIC MS NUMBER

MS21060L4K INDICATES: NUT, SELF-LOCKING, PLATE, TWO LUG, FLOATING, LOW HEIGHT, CRES; 125 KSI FTU, 450°F; DRY FILM LUBRICATION; .250-28 UNJF-3B; COUNTERSUNK OR DIMPLED HOLES.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

4. DESIGN AND USAGE LIMITATIONS: THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125 KSI BASED ON THE CROSS SECTION AREA AT THE BASIC ROOT DIAMETER OF THE THREADS. THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNAL THREADS. THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588. ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL 25027 SHALL BE USED.

USER ACTIVITIES
ARMY-CR, MI
NAVY-OS

REVIEWER ACTIVITIES
ARMY-AR
NAVY-SH
AIR FORCE-99
DLA-IS

APPROVED 23 JUL 64 REVISED (F) FOR CHANGES SEE PAGES 1,2 AND 3

PA NAVY-AS Other Cust ARMY-AV AIR FORCE-11	INTERNATIONAL INTEREST	TITLE NUT, SELF-LOCKING, PLATE, TWO LUG, FLOATING, LOW HEIGHT, CRES, 125 KSI FTU, 450° F & 800° F	MILITARY STANDARD MS21060
PROCUREMENT SPECIFICATION MIL-N-25027	SUPERSEDES NAS 686, NAS 1031 (IN PART)		PAGE 2 OF 3

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INTERCHANGEABILITY RELATIONSHIP

MS21060 NUTS CAN UNIVERSALLY REPLACE MAS686 AND MAS1031 NUTS OF LIKE MATERIAL, THREAD SIZE, LUBRICANT (DRY FILM OR SILVER PLATE) RIVET SPACING AND FASTENING METHOD (PLAIN RIVET HOLES, DIMPLED OR COUNTERSUNK RIVET HOLES OR PROJECTION WELDING), BUT THESE MAS686 AND MAS1031 NUTS CANNOT UNIVERSALLY REPLACE MS21060 NUTS.

INTERCHANGEABILITY TABLE

CANCELLED PART NUMBER	SUBSTITUTIVE PART NUMBER	CANCELLED PART NUMBER	SUBSTITUTIVE PART NUMBER	CANCELLED PART NUMBER	SUBSTITUTIVE PART NUMBER	CANCELLED PART NUMBER	SUBSTITUTIVE PART NUMBER
MAS686C04	MS21060-04	—	MS21060-3W	MAS1031C04	MS21060-04	MAS1031C4W	MS21060-4W
MAS686C04K	MS21060-04K	MAS686C3N	MS21060L3	MAS1031C04K	MS21060-04K	MAS1031C5	MS21060-5
—	MS21060-04W	MAS686C3NK	MS21060L3K	MAS1031C04W	MS21060-04W	MAS1031C5K	MS21060-5K
MAS686C04N	MS21060L04	—	MS21060L3N	MAS1031C06	MS21060-06	MAS1031C5W	MS21060-5W
MAS686C04NK	MS21060L04K	MAS686C4	MS21060-4	MAS1031C06K	MS21060-06K	MAS1031C6	MS21060-6
—	MS21060L04W	MAS686C4K	MS21060-4K	MAS1031C06W	MS21060-06W	MAS1031C6W	MS21060-6W
MAS686C06	MS21060-06	—	MS21060-4W	MAS1031C08	MS21060-08	MAS1031C7	MS21060-7
MAS686C06K	MS21060-06K	MAS686C4N	MS21060L4	MAS1031C08K	MS21060-08K	MAS1031C7W	MS21060-7W
—	MS21060-06W	MAS686C4NK	MS21060L4K	MAS1031C08W	MS21060-08W	MAS1031C8	MS21060-8
MAS686C06N	MS21060L06	—	MS21060L4N	MAS1031C3	MS21060-3	MAS1031C8W	MS21060-8W
MAS686C06NK	MS21060L06K	MAS686C5	MS21060-5	MAS1031C3K	MS21060-3K		
—	MS21060L06W	MAS686C5K	MS21060-5K	MAS1031C3N	MS21060-3N		
MAS686C08	MS21060-08	—	MS21060-5W	MAS1031C4	MS21060-4		
MAS686C08K	MS21060-08K	MAS686C5N	MS21060L5	MAS1031C4K	MS21060-4K		
—	MS21060-08W	MAS686C5NK	MS21060L5K				
MAS686C08N	MS21060L08	—	MS21060L5W				
MAS686C08NK	MS21060L08K	MAS686C6	MS21060-6				
—	MS21060L08W	—	MS21060-6W				
MAS686C3	MS21060-3	MAS686C6N	MS21060L6				
MAS686C3K	MS21060-3K	—	MS21060L6W				

USER ACTIVITIES
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PROCUREMENT SPECIFICATION MIL-N-25027	SUPERSEDES NAS 686, NAS 1031 (IN PART)		PAGE 3 OF 3

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